USB or SCSI?

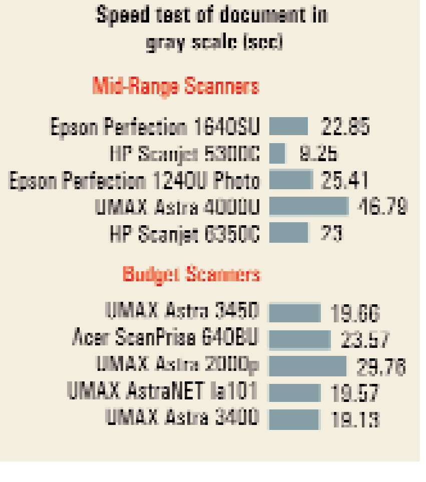
Almost all low-end scanners use a parallel port with a pass-through for a printer to also connect to the computer. This provides an inexpensive way to connect both the printer and the scanner to the computer. The downside to this is that speed is compromised and the other external Zip drives that need the parallel port cannot connect to the scanner. USB devices have been very successful in the market and you should opt for a USB connection over a parallel connection since it offers increased plug-n-play support, greater reliability and better flexibility.

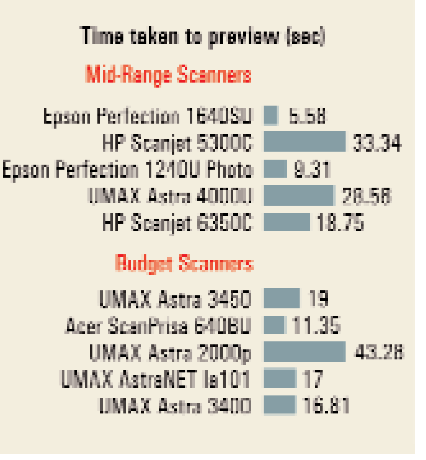
A high-end scanner usually comes with a SCSI connector. This is because SCSI connectors can handle a lot of data—a feature that is needed when dealing with large files. SCSI connectors usually ship with their own card and the SCSI connector of a scanner is likely to be incompatible with other SCSI cards.

If you are buying a low-end scanner with a resolution of less than 600 dpi, then opt for a USB device, while if you need a high-end scanner, a SCSI connector is a good choice.

It would be unfair to omit mentioning the Epson 1240U Photo, which wasn't far behind in any of the tests but couldn't make it to the ranks of the top three. Although it took more time than expected (84 seconds) for scanning the test photograph in full colour mode, it logged above-average results in all the other speed tests.

Overall, the HP ScanJet 5300C emerged





as the fastest scanner in the speed test and the quality of the scans was also quite impressive. The Epson Perfection 1640SU came in a close second.

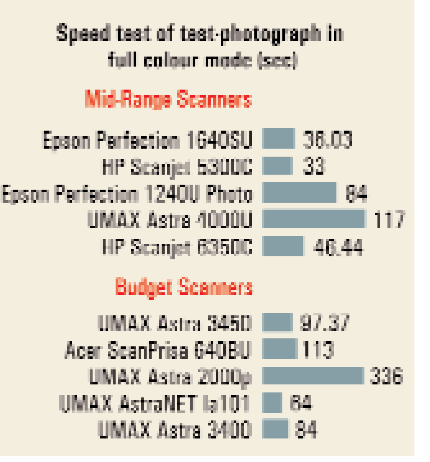
Image quality evaluation using IT8 card

This is one of the most comprehensive tests to evaluate the performance of imaging equipment. It was used to evaluate the ability of the scanners to differentiate between fine nuances in colour and luminosity.

To ascertain the scanner's ability to distinguish tonal resolution in gray scale areas, a set of 22 different blocks were observed to check for blending between adjacent blocks. This test determines the accuracy of the scanner in reproducing the various colour blocks in the IT8 card.

First, we checked if the scanner could accurately pick up the 22 different shades of colour. In order to log the most accurate values, we used the histogram function in Photoshop. We scanned the IT8 card onto Photoshop using the maximum optical resolution and bit-depth offered by the scanner in full colour mode. We selected an area that comprised 50 per cent each of two adjacent blocks and ascertained the standard deviation using Photoshop's histogram option. If the reading was greater than or equal to 2, then it implied that the scanner had differentiated between the two blocks. All the scanners but one passed this test by differentiating between adjacent blocks right up to block No 22. The exception was the Mustek 1200CUB, which could only detect differences among the first 21 blocks.

The next test performed using the IT8 card was the colour uniformity test. In this test, seven coloured blocks in the A row were selected in Photoshop. These were the lightest blocks in the row with very fine variations among them. Again the his-



togram function in Photoshop was used to determine the standard deviation of the red, green and blue colour values. The criterion was the same: a value of 2 or higher meant that the scanner had successfully detected the colour variations. Although none of the scanners scored below 2, many of them failed to pick up the green tones accurately.

The HP ScanJet 6350C recorded the

How Much Colour Depth?

Colour/bit depth is often wrongly associated with the number of colours a scanner can pick up. The bit depth is actually the number of levels of colour that a scanner can distinguish between. The extent of brightness values that a scanner captures (also called the dynamic range) depends upon two factors—the sensor and the analog-to-digital converter (ADC). For photos, the minimum depth needed is 24-bit, that is, 8-bits per colour or gray shade. However, even high-end 24-bit scanners often reduce the number of bits from 8 to 6. This means that the image does not have a 24-bit depth and so quality suffers. Also, the 6-bit colours tend to gather around the mid-tones, leading to deterioration in quality of the shadows and highlights. As the bit-depth increases, the output quality of the scanner should also increase. However, some manufacturers use a 24-bit scanner with a 10-bit ADC instead of an 8-bit ADC. This improves image quality to such an extent that it may even be better than a 30-bit scanner. However, you would need 30 bits or more for scanning slides or films.